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Analysis of birthweight and gestational age in antepartum stillbirths

***Jason Gardosi** *Senior Lecturer*, ***Theo Mul** *Research Fellow*, ***Max Mongelli** *Research Fellow*, †**David Fagan** *Senior Lecturer*

*Departments of *Obstetrics and Gynaecology, and †Histopathology, Queen's Medical Centre, Nottingham*

Objective To study the characteristics of birthweight and gestational age of third trimester fetal deaths which occurred before the onset of labour.

Design Review of computerised confidential perinatal mortality records. Data originated from the 1992 Trent Region Perinatal Mortality Survey.

Sample One hundred and forty-nine antepartum stillbirths of at least 24 weeks of gestation confirmed by early ultrasound scan. Congenital abnormalities and multiple pregnancies were excluded.

Main outcome measures Reported causes of stillbirth; weight-for-gestational age centiles based on a standard derived from normal pregnancies; pregnancy characteristics compared with the local maternity population.

Results Of 149 stillbirths, 83 (56%) were preterm and 66 were at term, and the majority (126; 85%) occurred from 31 weeks. Most of the deaths (97; 65%) were reported as 'unexplained' even though post-mortems had been carried out in 60% of all cases. Using a gestational age-specific fetal weight standard derived from normal, term live births, 41% of all cases of stillborn infants were small-for-gestational age (< 10th centile; OR 6·2; 95% CI 3·3–11·5); 39% of which had been classified as unexplained were small for gestational age (OR 5·6; 2·6–12·0). This excess of small stillbirths was most pronounced between 31 and 33 weeks, where the weights of 63% of all stillbirths and 72% of unexplained fetal deaths were < 10th centile. Overall, a higher proportion of preterm (< 37 weeks) than term stillbirths were small for gestational age: 53% *vs* 26% (OR 3·3; 1·6–6·5). However, at term there were also more subtle differences in weight deficit, with more fetuses with a weight between the 10th and 50th centiles than between 50th and 90th (36 *vs* 11; OR 3·3; 1·4–7·8). Mothers of pregnancies ending in stillbirth were similar in age, size, parity and ethnic group to mothers of live born babies, but were more likely to be smokers (37 *vs* 27%, OR 1·6; 1·2–2·3).

Conclusions Many stillborn babies are small for gestational age. In the absence of significant differences in physiological pregnancy characteristics, this is unlikely to be a constitutional smallness, but represents a preponderance of intrauterine growth restriction. For a full appreciation of the strength of this association, appropriate weight standards and classifications need to be applied in perinatal mortality surveys. Many antepartum stillbirths which are currently designated as unexplained may be avoidable if slow fetal growth could be recognised as a warning sign.

INTRODUCTION

In countries with low maternal mortality the major indicator of the quality of maternity care is the perinatal mortality rate. In the UK regular surveys have been conducted in Scotland[1] and Northern England[2] and the recently established Confidential Enquiry into Stillbirths and Neonatal Deaths in Infancy (CESDI) for England, Wales and Northern Ireland[3] has issued its first reports[4]. On the local as well as national level, such multidisciplinary enquiries can serve as an important audit of clinical performance. The events leading to death, and the result of a post-mortem when available, are the starting points for critical evaluation of whether the outcome was avoidable.

Correct classification of the cause of death is of primary importance, and two main methods have been proposed[3]. The Wigglesworth classification[5] is a pathophysiological method, with added subclassifications[6,7] to improve precision where autopsy reports are unavailable or sketchy. The Aberdeen classification is more clinical, includes maternal and obstetric factors and was used in the analysis of the 1958 British Perinatal Mortality Survey[8]. It too has been updated to include sub-classifications[9]. As the proponents of both systems

Correspondence: Dr J. Gardosi, Department of Obstetrics and Gynaecology, Queen's Medical Centre, Nottingham NG7 2 UH, UK.

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point out, the chief purpose of any classification is ultimately to help prevent perinatal deaths[5,8].

On the subject of antepartum deaths, however, neither classification method has been very successful in helping to determine the causes. The biggest contribution to perinatal mortality statistics is from stillbirths; of these, the so-called unexplained antepartum stillbirths amount to 68%[4], using the Wigglesworth classification[5]. Even when clinical classifications are used, regional reports have consistently shown that about two-thirds of normally formed, antepartum stillbirths are of unknown or unexplained causes[2,10,11]. This presents a problem when trying to improve clinical management, as fetal deaths are difficult to avoid if they are unexplained.

It is surprising that neither of the two classification methods has a separate category for fetal growth retardation. Intrauterine growth retardation (IUGR) is known to be associated with stillbirth in cases of fetal malformation, pre-eclampsia and multiple pregnancy, but it is also well recognised that in most cases, none of these conditions are present[12]. Such cases were included in a large category of 'deaths of uncertain origin', subdivided according to whether the baby weighed more or less than 2500 g[8]. Babies > 2500 g were considered mature and those weighing < 2500 g premature. This sub-classification was retained in the 1986 revision of the Aberdeen classification[9] and in the current large scale Confidential Enquiries[3,4].

Continued use in perinatal statistics of absolute weight categories, such as < 2500 g as the World Health Organization's definition of 'small baby', has been recommended[13] and retains usefulness when gestational dates are not known. However, there has also been an increasing awareness of the need to assess fetal size in relation to the duration of pregnancy. Ultrasound scans in the first half of pregnancy can establish gestational age better than even a 'certain' menstrual history[14], and routine dating scans have allowed the definition of more precise birthweight standards[15]. But the weights of premature births are heavily skewed as they represent an abnormal population. In contrast, ultrasound studies have been able to delineate fetal weight norms in low risk pregnancies with normal outcome[16]. We applied such a standard to investigate the association between low weight for gestation and stillbirth in a database of intrauterine deaths which occurred before the onset of labour.

METHODS

Stillbirth datafiles

The data for antenatal stillbirths originates from the 1992 Trent Regional Epidemiological Survey of Perinatal Mortality. The computerised perinatal mortality data were obtained from the Trent Health Authority co-ordinators in anonymised form. There were a total of 62,142 deliveries in Trent Region during 1992, and 272 stillbirths after 24 weeks of gestation, which is equivalent to a rate of 4·38 per thousand. This compares with a rate of 4·25 per thousand stillbirths of 689,656 deliveries in England and Wales during 1992. The degree of maceration was not usually stated, but comment was made as to whether the death was considered to have occurred before or after the onset of labour. We excluded intrapartum deaths (n = 20) and cases which had not been categorised as to whether fetal death occurred antenatally or intrapartum (n = 17), as well as multiple pregnancies (n = 14) and congenital abnormalities (n = 10). A further 28 cases were excluded because gestation dates had not been confirmed by ultrasound before 21 weeks and 34 cases where data were incomplete in other respects. This left a total of 149 computer files of singleton, nonmalformed antepartum fetal deaths from 24 weeks of gestational age for analysis.

Gestational age

To determine the duration of pregnancy, the gestational age at delivery was calculated in days. Estimates of time intervals between fetal demise and delivery were not available, and the degree of maceration was often not recorded. We estimated that in the third trimester, when the mother's perception of fetal movements is usually well established, the average time interval between fetal death and spontaneous or induced delivery is 48 hours. Such an average delay is consistent with histopathological assessments[17], and with post-mortem examinations in our own unit, where the time of death is considered to have occurred within the preceding 72 hours in over 90% of cases. An average correction for the death-to-delivery delay was made by subtracting two days from the length of pregnancy in each case before calculating the gestational-age specific weight centile.

Stillbirth weight

The weights of the stillborn babies in the Trent database were those recorded by labour ward staff after delivery. To test the accuracy of such weight records, we examined a detailed subset of stillbirths in Nottingham which also had a record of the 'dry' weights, determined at the outset of a post-mortem examination. There was a high correlation between paired weight measurements of macerated stillbirths (n = 39, r = 0·986; weight at delivery on average +28·5 g or 1·4% higher than weight recorded at post-mortem), as well as nonmacerated stillbirths (n = 15, r = 0·997, mean difference +21 g or 1·0%).

Weight standards

The ultrasound derived fetal weight standard of Hadlock *et al.*[16] was used to calculate gestational age-specific centiles. It is based on Hadlock's own fetal weight formula using four parameters (fetal head circumference, biparietal diameter, abdominal circumference and femur length) which has been shown to perform well at term as well as preterm gestations and across a range of different weights, with an overall systematic error of −1·2%[18]. The rationale for a fetal, rather than a neonatal weight standard is that it is derived from a population of pregnancies which went to term and had a normal outcome. It shows a normal distribution of weight throughout gestation[16]. In contrast, neonatal weights in the preterm period are heavily skewed[15] as they represent an abnormal population, which limits its use as a standard.

Comparative analysis has shown the Hadlock fetal weight curve to have close similarity to several other recently published curves[19,20]. It was chosen because 1. it shows minimal systematic weight estimation error and close similarity between average fetal weight and birthweight in its study population (3619 g *vs* 3598 g; +0·5%) and 2. it is derived from a sample which has a similar 40 week birthweight as that observed in our own population[15] (3598 *vs* 3531 g or −1·9%). The difference is even smaller when using coefficients to adjust for pregnancy characteristics[21,22]: the average 40 week birthweight for low risk pregnancies in Nottingham (average maternal height and weight, para 1, Anglo-European ethnic group, nonsmoking) is 3586 g (ie, almost identical to the 3598 g obtained in a similar low risk population of Hadlock's ultrasound study[16]).

Population characteristics

The maternal characteristics of the stillbirth cohort were compared with those from the Nottingham computerised obstetric database[15], updated to 1995. This consisted of 49,657 consecutive, ultrasound dated, complete records of singleton pregnancies between 1988 and 1995 which resulted in a live birth.

Data analysis

A software program was written to calculate gestation-specific weight centiles. Statistical analysis was carried out with SPSS for Windows (Version 6, SPSS Inc. , Chicago, Illinois, USA); odds ratios and confidence limits were calculated using statistical software by CentorSoft, Richmond, Virginia, USA.

RESULTS

The reported causes of antepartum fetal deaths in the dataset of 149 stillbirths, which excluded multiple pregnancies, congenital abnormalities and intrapartum deaths, are listed in Table 1. In the majority of cases (65%), no cause had been ascertained. A post-mortem had been performed in 89 cases (60%). No cause of death was identified in 61/89 cases (69%) which had a post-mortem, and in 36/60 cases (60%) which did not have a post-mortem (OR 1·5; 0·7–2·9). More than half of all stillbirths (83 of 149; 56%) occurred during the preterm period (24 to 36 completed weeks), which compares with 7% of preterm live births in our general population. Most stillbirths (126/149, 85%) had a gestational age of at least 31 weeks.

In Table 2 pregnancy characteristics of stillbirths are compared with an unselected dataset of 49,657 live

Table 1. Reported causes of death in group of 149 stillbirths in Trent Region in 1992.

Cause of stillbirth	*n*	(%)
Unexplained	97	65
Abruption	22	15
Growth retardation	12	8
Pre-eclampsia	6	4
Cord accident	6	4
Infection	2	1
Miscellaneous	4	3
TOTAL	149	100%

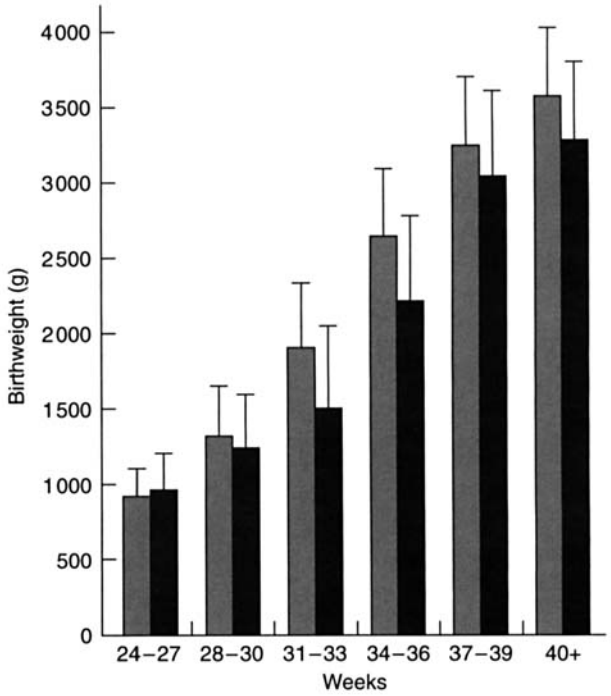

Fig. 1. Mean SD birthweight of live born infants in Nottingham, 1988–1995, and stillborn infants in the Trent Region, 1992. ■ = stillbirths; ■ = livebirths.

Table 2. Characteristics of pregnancies resulting in stillbirths in Trent sample (n = 149) compared with Nottingham computerised database of live births (n = 49,657). Values are given as n (%) or mean [SD]. NS = not significant.

	Stillbirths (n = 149)	Live births	OR (95% CI)	P
Sex ratio		(coded: 48509)		
Males	83 (55·7)	24,846 (51·2)	1·20 (0·86–1·65)	
Male: females	1·26	1·05		
Maternal age (years)				
Mean [SD]	27·2 [5·5]	26·5 [5·2]		NS*
< 20	12 (8·1)	4335 (8·7)	0·89 (0·50–1·60)	
35+	15 (10·1)	3437 (6·9)	1·48 (0·87–2·50)	
Maternal height (cm)				
Mean [SD]	161·0 [6·9]	162·4 [6·4]		< 0·05*
< 152 cm (< 5 ft)	9 (6·0)	1774 (3·6)	1·69 (0·87–3·28)	
Ethnic group		(coded: 47249)		
Anglo-European	140 (94·0)	43324 (91·6)	1·41 (0·73–2·73)	
Indian/Pakistani	4 (2·7)	2196 (4·6)	0·57 (0·22–1·47)	
Afro-Caribbean	3 (2·0)	1210 (2·6)	0·78 (0·26–2·32)	
Other	2 (1·3)	519 (1·1)	1·23 (0·33–4·51)	
Parity		(coded: 47166)		
Primiparae	69 (46·3)	20620 (43·7)	1·11 (0·81–1·53)	
Smokers		(coded: 46616)		
Any	55 (36·9)	12399 (26·6)	1·61 (1·16–2·25)	
10+ / day	25 (16·8)	7187 (15·4)	1·27 (0·82–1·96)	
20+ / day	3 (2·0)	1428 (3·1)	0·76 (0·26–2·29)	

*Mann Whitney U test.

births in the Nottingham population. There were no significant differences in maternal physiological characteristics, with the exception of a clinically small difference in maternal height. However, mothers with stillborn babies were more likely to have smoked during pregnancy. This overall higher rate of smoking was a feature of pregnancies with preterm (34/83, 41%) more than term stillbirths (21/66, 32%; OR 1·5, CI 0·8–2·9).

The average weights of stillbirths were lower than those of live born babies in Nottingham at each gestational age category except that from 24 to 27 weeks (Fig. 1). The weight distribution of all stillbirths against the fetal weight-for-gestational age standard is shown in Fig. 2. Overall, 61 of the 149 stillbirths (41%) were small for gestational age, defined as < 10th centile (OR 6·2; 3·3–11·5). A higher proportion of preterm (< 37 weeks) than term stillbirths were small for gestational age (SGA): 53% *vs* 26% (OR 3·3; 1·6–6·5). At term, however, there were also more fetuses with a weight between the 10th and 50th centiles than between 50th and 90th (36 *vs* 11; OR 3·3; 95% CI 1·4–7·8). Altogether, 53/66 term stillbirths (80%) had a weight below the median for their gestational age, compared with the expected 50% (OR 4·1; 1·9–8·7; $P < 0·001$).

Of the 97 unexplained deaths, 38 (39%) were SGA (OR 5·6; 2·6–12·0), and 56% of the unexplained stillbirths in the preterm period were SGA. The largest proportion of SGA stillbirths was between 31 and 33 weeks, where 66% were below the 10th centile and 56% were below the 5th centile of the fetal weight standard. Further analysis of this 31–33 week gestation period is shown in Fig. 3. Neonatal weights are compared with

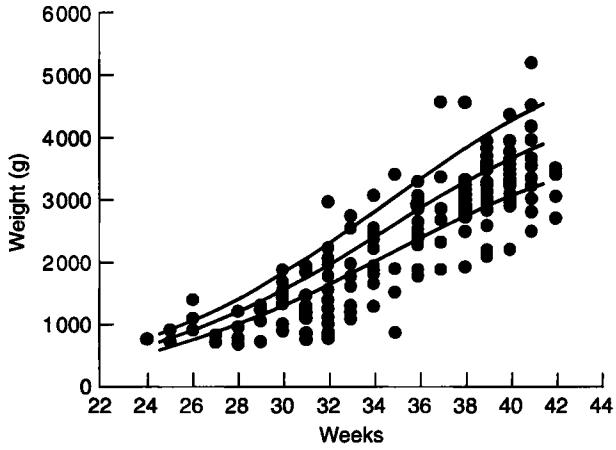

Fig. 2. Birthweights of 149 singleton, nonmalformed stillbirths in Trent Region in 1992, plotted against 10th, 50th and 90th centile lines of the local fetal weight standard. Gestational age based on ultrasound and adjusted for delay to delivery.

the fetal weight distribution of normal term pregnancies, demonstrating a negative skewness of the weights of babies that are born at this gestation. The weights of stillbirths (n = 32) in this gestational age category are also shown. There is an apparent bimodal distribution, with most cases falling below the 10th fetal weight centile value for 32 weeks, 1621 g[16]. The causes of death for these cases are listed in Table 3. Twenty of the 32 stillbirths (63%) fall below the tenth centile cut-off, including 13/18 (72%) of the unexplained deaths. Most of these (12/18 cases, 67%) are even below the 5th centile limit for fetal weight at 32 weeks (1545 g).

DISCUSSION

This analysis shows that many antepartum stillbirths are smaller than expected for their gestational age. The effect becomes apparent by the application of an ultrasound derived fetal weight standard, and seems most marked in the preterm period. The results suggest a strong association between fetal death and a preceding slow growth rate.

Small for gestational age is not synonymous with being growth retarded[23]. Apart from pathological growth failure, babies can be SGA because of individual genetic or constitutional factors. In part, such variation is predictable, and is influenced by maternal size, parity, and ethnic group, as well as the gender of the baby[21,22]. These variables were not found to be markedly different between live birth and stillbirth groups (Table 2). We were not able to adjust the weight standard for each individual pregnancy as no data were available for maternal weight which is one of the strongest variables affecting birthweight and fetal weight[22]. In individual cases, smallness can be attributed to normal variation and some babies with constitutionally slow growth may perhaps also have an increased predisposition to intrauterine death; but the lower weight centiles of stillborn babies as a group is more likely to reflect a pathological association (ie, growth retardation). One reason for this could be smoking, a feature of more pregnancies that ended with stillbirth, which agrees with previous observations[24]. This adds to the argument that smoking should be regarded as an important marker for high risk pregnancies which require increased surveillance. However, while there is a quantifiable, dose dependent effect of

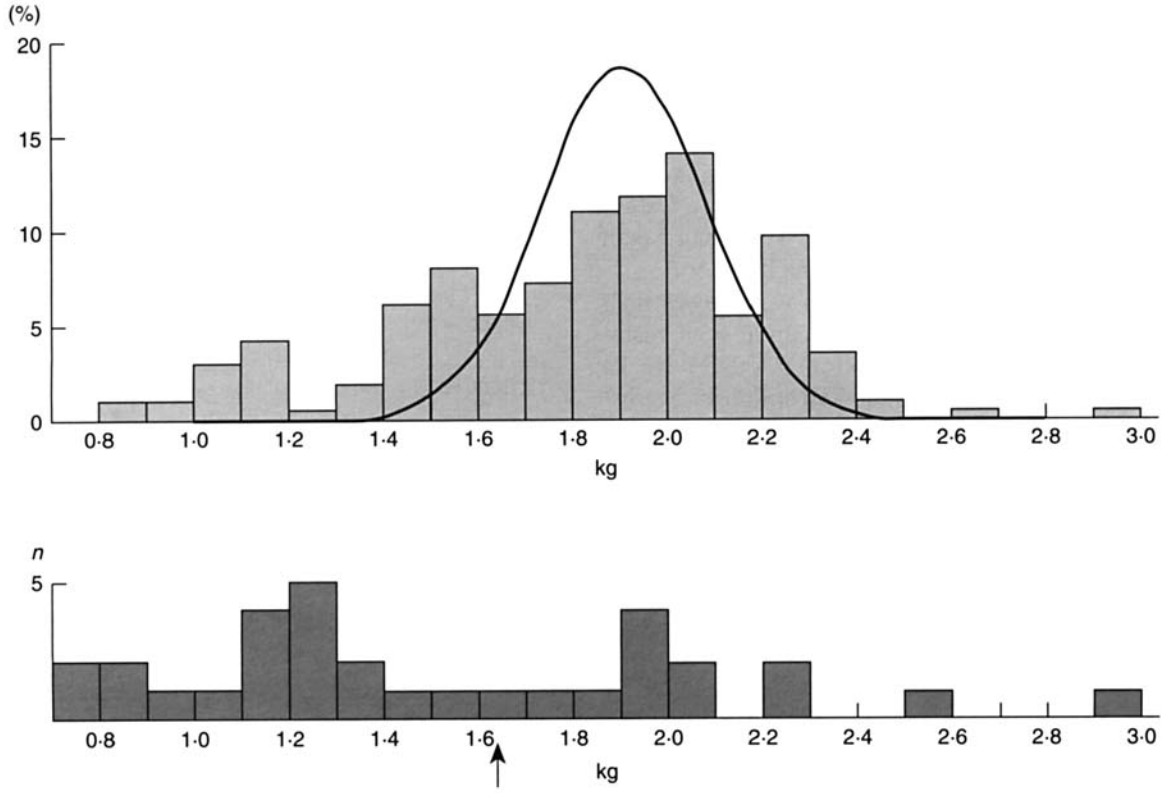

Fig. 3. Distribution of neonatal weights in Nottingham (top), and stillbirth birthweights in the Trent survey (bottom) at 31 to 33 weeks of gestation. The bell-shaped curve shows the normal fetal weight distribution at 32 weeks[16], and the arrow indicates the 10th fetal weight centile limit, 1621 g, at that gestation.

Table 3. Weight-for-gestational age categories of stillbirths at 31–33 weeks in Trent Region, 1992. Values are given as *n*.

	< 10th	≥ 10th	TOTAL
Unexplained	13	5	18
Abruption	4	5	9
IUGR	3	—	3
Pre-eclampsia	—	1	1
Infection	—	1	1
TOTAL	20	12	32

smoking on birthweight[22], there was a similar proportion of small for gestational age stillborn infants among mothers who smoked as among those who did not. Most stillbirths occurred in pregnancies where the mother did not smoke.

It is important to make every attempt to establish the cause of death in each individual case. One needs to consider physiological characteristics, pregnancy events, and preceding clinical or ultrasound observations where available. Awareness that an infant was small for gestational age should heighten the pathologist's level of suspicion and indicate a targeted examination of individual organs such as the thymus, adrenal or placenta. Even so, changes may be subtle and fail to reveal any obvious abnormality; chronic deprivation may cause relative organ changes only[12], so that even the most thorough post-mortem may not reveal a cause.

In this series most stillbirths were reported as being of unexplained causes, regardless of whether a post-mortem had been performed. Neither the pathophysiological[5] nor the obstetric[8,9] classification recognises growth retardation as a separate diagnostic category. The recommended subgrouping according to absolute weight categories, such as < 2500 or < 1500 g[3,4,9], is not helpful when considering antepartum deaths because it obscures the distinction between smallness due to prematurity and smallness due to growth retardation. A further problem is the widespread use in post-mortem reports of a weight-for gestation reference standard which is based on a database of other stillbirths[25]. The use of weights of dead infants as reference will obscure the association between small for gestational age and fetal death (Fig. 4). These stillbirth reference curves seem even lower than the distribution of weights in our dataset, presumably because we excluded multiple pregnancies, malformations, and cases with uncertain gestational dates.

Even if live birth weights are used for reference, the wrong standards are applied to premature deliveries. Weight-for-gestational age data derived from preterm neonates cannot be regarded as normal and they have a wide, negatively skewed distribution[15] which will affect the cut-off limit for defining small for gestational age. The inappropriate use of preterm neonatal weights as a

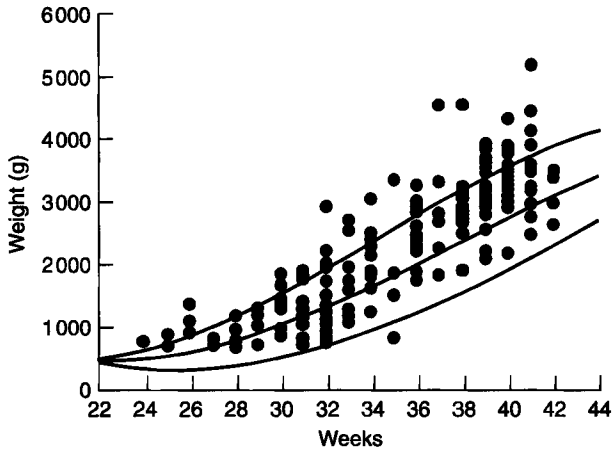

Fig. 4. Scatterplot of weights of 149 stillbirths, against smoothed 10th, 50th and 90th centile lines of the Rhode Island postmortem weight standard[25].

'standard' may explain why the largest single category of antepartum, singleton, normally formed stillbirths remains that due to unexplained or unknown causes, even in studies where growth retardation had been given its own diagnostic category[26,27].

The proportion of stillborn infants that are found to be small for gestational age when applying a normal population fetal weight standard is considerable. At preterm gestations, 53% of stillbirths weigh less than the 10th centile (ie, an excess of 43% over that expected in normal pregnancies). This implies that fetuses who are small for gestational age have a significantly higher risk of intrauterine death. At term, stillbirth weights were also lighter but the effect was more subtle: there was an excess of recorded weights in the 10th to 50th centile range, compared with 50th to 90th centile (Fig. 2). These babies are not small for gestational age by conventional definition, but may still have failed to reach their own growth potential. Perhaps in these late fetal deaths the causes of slower growth were less severe but more prolonged.

Relative smallness, whether constitutional or due to growth retardation, is unlikely to be the immediate reason for fetal demise, but provides important information for clinicians who are concerned with prospective management and prevention. Some paediatricians and obstetricians have come to doubt the significance of growth retardation among infants born at term[28,29] and their arguments seem to be supported by evidence that these babies display catch-up growth[30-32]. But these babies may only be the lucky survivors of an unfavourable intrauterine environment. Fetuses in the lower weight ranges are at increased risk of intrauterine death with advancing gestational age[33].

Awareness of a strong link between small for gestational age and stillbirth is important for local evaluation

of obstetric performance as well as for large scale perinatal mortality surveys. The preponderant label of unexplained antepartum stillbirths in perinatal statistics may be misleading if it is taken to imply that these deaths are unavoidable. The excess in small for gestational age babies is under-reported because the classification systems and weight standards in use are inappropriate for evaluating stillbirths. The majority of stillbirths in this study (85%) occurred from 31 weeks, when live birth survival in most units in this country is better than 95%. The question must arise whether early recognition would not result in prevention of some of these deaths. We suggest that renewed emphasis should be placed on improving strategies for antenatal detection and surveillance of babies with slow intrauterine growth.

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